

TOLE BI, Kazakhstan

WMO: 382220

Lat: 43.700N

Lon: 73.783E

Elev: 1496

StdP: 13.92

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-4.7	-15.3	2.5	-11.2	-8.6	3.7	-4.2	13.0	26.1	10.3	27.8	1.1	180	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	97.5	67.5	94.7	66.8	92.2	65.8	70.6	90.6	68.9	89.1	67.5	87.6	4.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	94.0	78.9	62.1	88.1	76.9	60.4	83.0	75.9	35.4	90.8	34.0	89.2	32.8	87.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	12.4	9.8	-16.9	102.3	8.9	2.2	-23.4	103.8	-28.6	105.1	-33.6	106.3	-40.1	107.9		
(5)			-17.0	74.2	8.6	2.5	-23.2	76.0	-28.3	77.5	-33.1	78.9	-39.4	80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	21.0	26.4	44.0	55.8	65.3	74.1	77.0	74.7	64.8	51.5	37.3	24.2
	DBStd	21.59	12.01	12.68	10.25	7.85	6.48	5.35	4.93	5.05	7.32	8.14	10.55	11.57
	HDD50	3102	899	661	233	35	1	0	0	0	1	79	393	800
	HDD65	6074	1364	1081	650	285	75	4	1	2	94	424	830	1264
	CDD50	3642	0	1	48	209	477	723	836	766	444	125	13	0
	CDD65	1137	0	0	0	9	85	277	371	303	87	5	0	0
	CDH74	15352	0	0	26	237	1305	3571	4900	3931	1257	121	4	0
	CDH80	7300	0	0	4	52	447	1721	2582	1983	486	25	0	0
(14) Wind	WSAvg	3.1	1.9	2.7	3.8	4.2	4.2	3.7	3.4	3.5	3.1	2.5	2.2	1.8
(15) Precipitation	PrecAvg	12.1	1.0	0.9	1.2	1.5	1.5	0.9	0.4	0.3	0.4	1.3	1.2	1.5
	PrecMax	20.2	3.0	2.4	2.8	3.8	5.0	3.2	1.4	1.3	1.1	3.3	2.2	11.7
	PrecMin	8.1	0.1	0.0	0.4	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.2	0.0
	PrecStd	2.7	0.6	0.6	0.7	0.9	1.2	0.7	0.4	0.4	0.3	0.9	0.6	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	59.0	77.5	85.6	92.7	98.7	102.5	99.9	93.7	83.5	70.2	51.1
		MCWB	42.8	47.5	55.0	60.4	64.4	68.5	68.9	67.8	65.7	59.2	53.8	43.5
	2%	DB	42.9	51.1	71.0	80.5	88.5	95.2	98.2	96.3	89.4	77.1	62.1	45.4
		MCWB	37.7	42.9	53.4	59.0	63.2	66.9	68.3	66.8	62.7	56.8	51.5	40.1
	5%	DB	38.7	45.6	66.0	76.0	85.0	92.4	95.1	93.6	85.9	72.4	56.5	41.1
		MCWB	35.0	39.9	51.7	57.4	62.4	65.9	67.1	66.0	61.1	54.5	47.8	37.3
	10%	DB	35.3	40.9	60.8	71.8	81.5	89.4	92.1	90.5	81.7	67.6	51.8	38.0
		MCWB	32.8	36.9	49.4	55.5	61.0	65.1	66.4	64.8	59.6	52.3	45.2	35.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.0	48.0	58.0	64.4	68.5	72.2	73.1	71.8	67.3	60.4	56.7	44.9
		MCDB	49.6	58.1	72.5	77.3	85.6	90.0	91.9	90.6	89.4	77.3	67.4	50.2
	2%	WB	38.4	43.4	54.6	61.5	66.1	69.8	70.9	69.1	64.3	58.0	52.2	40.6
		MCDB	42.0	49.9	68.2	75.4	82.9	89.5	91.4	90.4	85.1	74.1	60.4	44.2
	5%	WB	35.2	40.0	52.3	59.3	64.3	68.0	69.2	67.6	62.3	55.9	48.7	37.6
		MCDB	38.2	45.1	63.8	72.6	80.2	87.7	89.3	89.6	82.8	69.4	54.9	40.9
	10%	WB	32.8	36.9	49.8	57.0	62.6	66.6	67.8	66.1	60.5	53.5	45.6	35.1
		MCDB	35.3	40.6	59.8	69.2	77.9	85.1	87.7	87.4	79.5	65.6	51.4	37.8
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	18.1	22.6	23.8	26.3	26.7	27.5	28.4	28.8	25.5	19.3	16.3
		MCDBR	19.1	22.8	31.2	32.5	32.7	31.4	32.2	33.1	33.6	33.2	26.8	19.9
	5% WB	MCWBR	14.5	15.8	16.4	15.1	13.1	11.5	11.6	12.5	13.9	16.2	16.7	15.0
		MCWBR	16.7	21.7	28.0	27.4	27.9	28.3	28.3	30.2	31.8	28.6	23.1	18.5
(40) Clear-Sky Solar Irradiance	taub		0.357	0.397	0.443	0.466	0.450	0.446	0.437	0.405	0.386	0.416	0.399	0.352
		taud	2.085	2.025	1.978	1.967	2.073	2.129	2.168	2.239	2.239	2.062	2.046	2.088
	Ebn at Noon		230	242	247	253	262	263	264	269	264	232	210	217
		Edh at Noon	33	42	50	55	51	48	46	41	38	40	35	30
(44) All-Sky Solar Radiation	RadAvg		587	881	1243	1679	2136	2295	2280	2087	1641	1037	620	461
	RadStd		65	94	94	142	135	110	117	86	65	92	58	42

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-2.27	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page